

1. Introduction

Purpose

The purpose of this document is to define the functional and non-functional requirements for the CMS. It is intended for project managers, developers, and college administrators.

Project Scope

The CMS is an integrated platform designed to digitize administrative tasks, student records, and academic tracking. It replaces paper-based systems with a centralized database to ensure data integrity and real-time accessibility.

2. Overall Description

Product Perspective

The system operates as a web-based application with a centralized database. It may interface with external systems like bank APIs for fee payments and SMTP servers for email notifications.

User Classes and Characteristics

Administrator: Full access to system configuration, user management, and financial reports.

Faculty: Access to academic modules, attendance marking, and grade entry.

Student: Access to personal profiles, timetables, and fee status.

3. Operating Environment

Client: Modern web browsers (Chrome, Firefox, Safari).

Server: Linux-based OS (Ubuntu/CentOS).

Database: RDBMS like MySQL or PostgreSQL.

4. System Features (Functional Requirements)

User Authentication Module

REQ-1: Users must log in with a unique ID and encrypted password.

REQ-2: The system must support Password Recovery via registered email.

Student Information Management

REQ-3: Admins must be able to Create, Read, Update, and Delete (CRUD) student records.

REQ-4: The system shall generate unique Enrollment Numbers automatically.

Attendance & Academic Tracking

REQ-5: Faculty shall record daily attendance.

REQ-6: The system shall calculate the attendance percentage. If A is the number of sessions attended and T is the total sessions:

$$\text{Attendance \%} = (A/t) * 100$$

Examination & Result Module

REQ-7: The system must allow the entry of internal and external marks.

REQ-8: Automated CGPA calculation based on credit weights.

5. Non-Functional Requirements

Security

All data transmissions must be encrypted using SSL/TLS.

Sensitive data like passwords must be hashed (e.g., using BCrypt).

Reliability

The system should have a Mean Time Between Failures (MTBF) of at least 3,000 hours.

Weekly automated backups of the database are mandatory.

Performance

The system shall respond to user queries within 2 seconds under normal load.

The system must support at least 500 concurrent users.

6. Data Requirements (Entity Relationship)

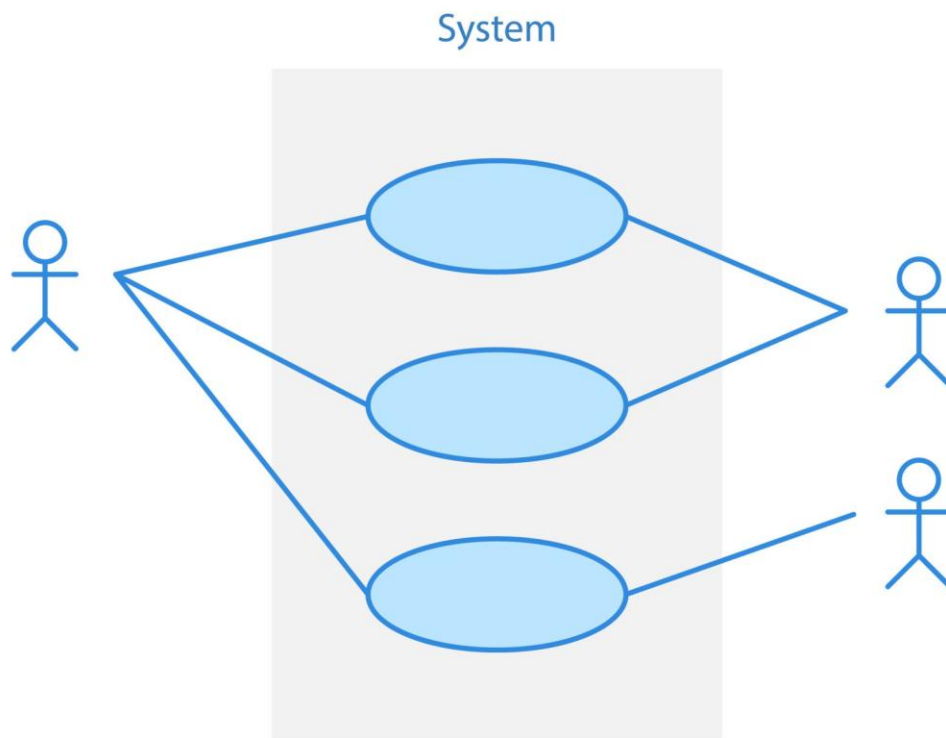
To maintain data integrity, the following primary relationships must be established:

One-to-Many: One Department to many Students.

Many-to-Many: Many Students to many Courses (via an Enrollment table).

7. Use Case Diagram (Functional Scope)

This diagram defines the "who" and "what." it shows how the Student and Registrar interact with the enrollment features.



Actors: Student, Registrar, Database.

Key Actions: Login, Search Courses, Check Eligibility, Register, Pay Fees.

8.Conclusion

The College Management System (CMS) serves as a vital digital backbone for modern educational infrastructure. By integrating student enrollment, academic tracking, and administrative workflows into a single unified platform, the system effectively eliminates manual redundancies and enhances data integrity. Guided by the requirements outlined in this SRS, the development team can ensure a scalable, secure, and user-centric application that empowers students, faculty, and administrators with real-time insights.